

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 104 — Past Questions & Materials
by Renance

Course: PHY 104
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 100

This study pack was compiled and restructured by Renance for PHY 104 students at Federal University of Petroleum Resources, Effurun. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

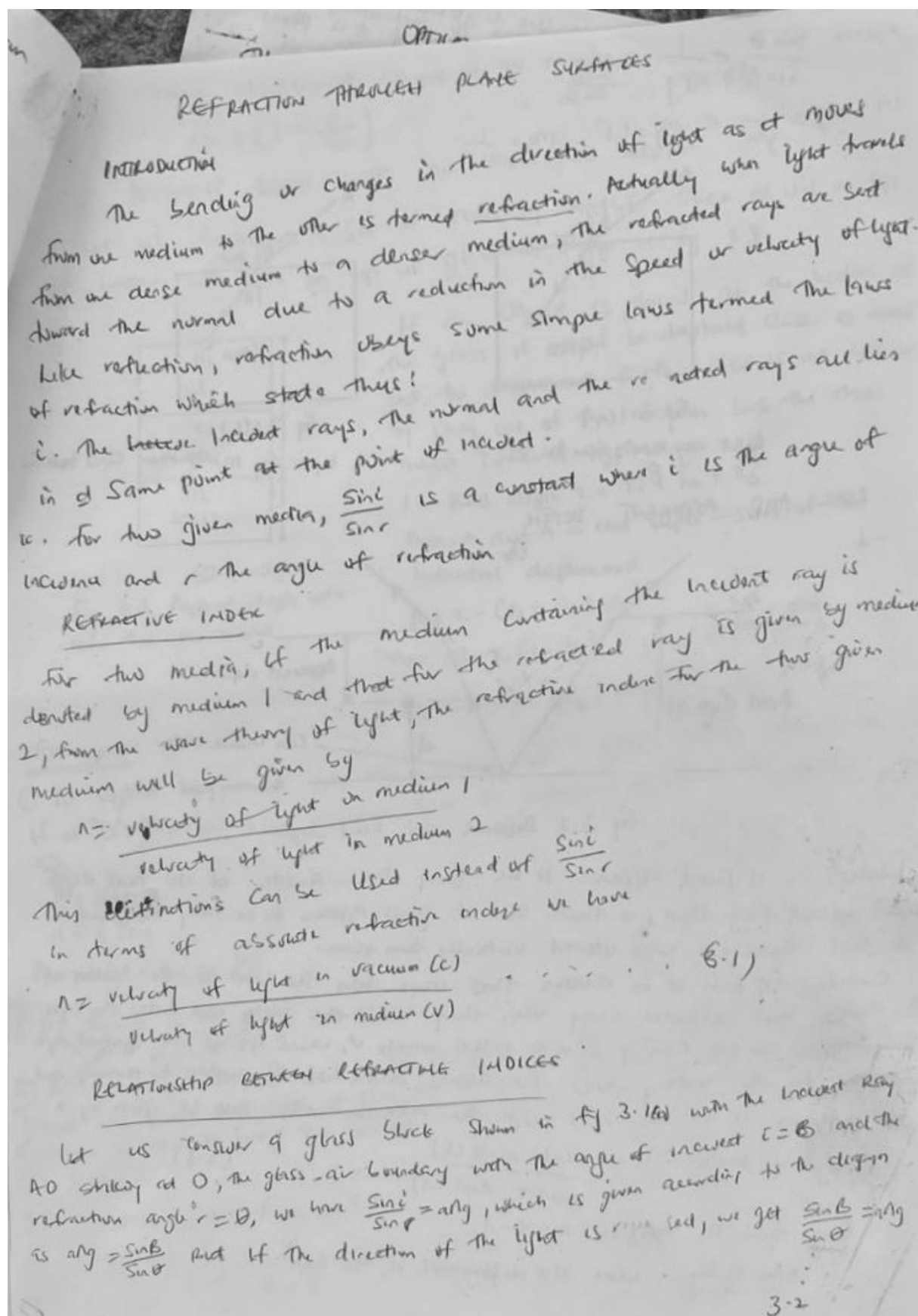
Study online: renance.app
Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 1 — PHY 104

Level 100 • Second Semester • Handout

This source paper is a scanned document; its pages are reproduced below.



Since $\frac{\sin \theta}{\sin \theta'} = \frac{1}{n_g} \Rightarrow \frac{\sin \theta}{\sin \theta'}$

$\therefore n_g = \frac{1}{\sin \theta'}$ If $n_g = 1.5$ $\sin \theta' = \frac{1}{1.5}$

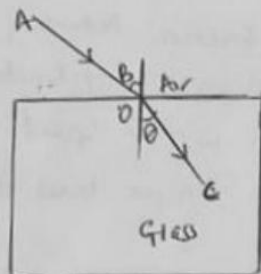


Fig 3-1(a)

Light ray moving from Air into Glass

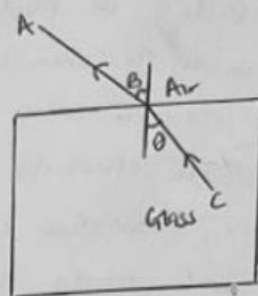


Fig 3-1(b)

Light ray moving from Glass into Air

REAL AND APPARENT DEPTH

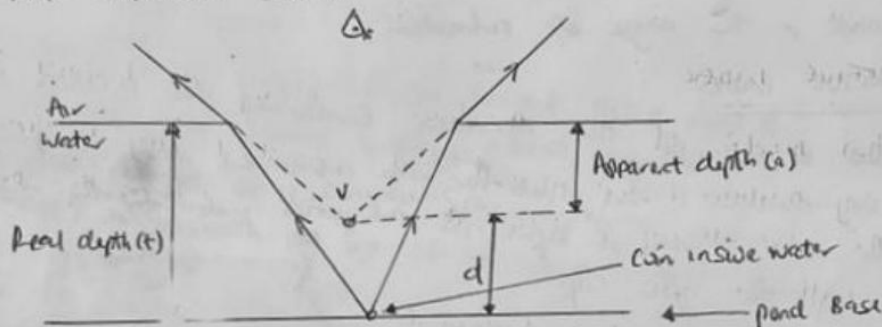


Fig 3-2 Apparent and Real Depth.

Water in a pond appears to be only three-quarter of its real depth when viewed from above, a thick slab of glass appears to be only two-third of its real thickness when viewed vertically from above.

Consider Fig 3-2 it is obvious that rays from the coin at the bottom of the water are refracted away when they leave the water and enter the eyes. They appear to be coming from a virtual image V , which is at an apparent depth a below the water, while the actual depth of the bottom t , remains and is referred to as the real depth, the refractive index will be given by

$$\text{Refractive index } (n) = \frac{\text{real depth } (t)}{\text{apparent depth } (a)} \quad (2.3)$$

But from the diagram $a = t - d$

$\therefore n = \frac{t}{t-d}$ where $d = \text{displacement of the coin}$

making displacement subject of the formula

$$dz \approx t \left(1 - \frac{1}{n_w}\right) \quad \dots \dots \dots (3.4)$$

APPARENT DEPTH WITH MORE THAN TWO MEDIA

Let us consider a set of refracting media such as oil on top of water and both on top of glass as shown in fig 3.3

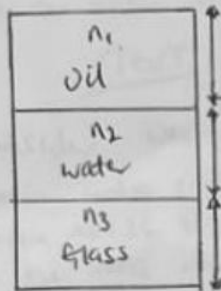


Fig 3.3 Apparent depth with more than two media

If an object is placed at the bottom of the glass, it would be displaced closer as viewed but the displacement in this case is not determined by only one of the medium but the three media combined together.

$$\text{i.e. Real depth } x = x_1 + x_2 + x_3$$

Apparent depth $A = \text{real depth} - \text{sum of the individual displacements}$

$$A = x - (d_1 + d_2 + d_3)$$

$$\text{where } d_1 = x_1 \left(1 - \frac{1}{n_1}\right) \quad d_2 = x_2 \left(1 - \frac{1}{n_2}\right)$$

$$d_3 = x_3 \left(1 - \frac{1}{n_3}\right) \text{ etc.}$$

Examples

- ① An object is placed at the bottom of a glass block 18 cm thick. If it is displaced 6 cm upwards, find the refractive index of the glass.

Soln

$$t = 18 \text{ cm}$$

$$d = 6 \text{ cm}$$

$$\text{Recall } d = t \left(1 - \frac{1}{n}\right)$$

$$n = \frac{t}{t-d} = \frac{18-0}{18-6} = \frac{18}{12} = 1.5$$

- ② A microscope was focused on a scratch at the bottom of a jar when a liquid was poured to a depth of 8 cm. The microscope was a distance of 1.4 cm to bring the scratch back into focus. find the refractive index of the liquid?

Soln $t = 8 \text{ cm}, d = 1.4 \text{ cm}, n = ?$

$$\text{Using } n = \frac{t}{t-d} = \frac{8}{8-1.4} = 1.212 \approx 1.2$$

Since v is negative, ~~then~~ the image is virtual. The linear magnification will be gotten from $M = -\frac{v}{u} = -\frac{(-24)}{40} = 0.6$

Since M is positive but less than 1, thus the image is erect but diminished.

- ③ A mirror formed an erect image one-fifth the size of an object placed 15cm in front of it. If the mirror is spherical, what kind is it, and what is its radius of curvature.

Soln

$$u = 15 \text{ cm}$$

$$m = 1/5$$

$$f = ?$$

$$r = ?$$

$$\text{Using } m = -\frac{v}{u}$$

$$\frac{1}{5} = -\frac{v}{15}$$

$$v = -3 \text{ cm}$$

Since $m = 1/5$, and $v = -3 \text{ cm}$ then the image is virtual, diminished and erect thus the mirror must be convex.

Using the mirror equation $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\frac{1}{f} = \frac{1}{15} + \frac{1}{-3} = -\frac{4}{15}$$

$$f = -\frac{15}{4} \text{ cm } (-3.75 \text{ cm}) \quad r =$$

- ④ If an object is 30cm in front of a convex mirror that has a focal length of 60cm, how far behind the mirror will the image appear to an observer? How tall will the image be.

Solution

$$u = 30 \text{ cm}$$

$$f = -60 \text{ cm (convex)}$$

$$v = ? \quad m = ?$$

$$\text{Using } \frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

$$v = \frac{uf}{u-f} = \frac{30(-60)}{30-(-60)} = -\frac{1800}{90} = -20 \text{ cm}$$

i.e. The image will be formed 20cm behind

① The height of the image is gotten from the magnification

$$M = -\frac{v}{u} = -\frac{(-20)}{30} = \frac{2}{3}$$

Thus the image will be $2/3$ the height of the object.

$$A = \text{total thickness} - \text{total displacement}$$

$$= (8 + 6 + 10) - 7.9103$$

$$= (24 - 7.9103) \text{ cm}$$

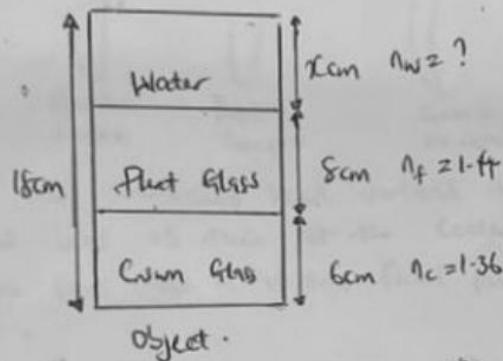
$$= 16.089 \text{ cm.}$$

While the apparent position of the object from all the above will correspond to the value 7.9103 which is the resultant of placement.

Q6C1 to be immediately

Consider water on top of flint glass of thickness 8cm, refractive index 1.44, resting on top of a crown glass 6cm thick and refractive index $n_c = 1.36$. If the displacement due to the combination is 6cm from below with the total thickness being 18cm. Compute the refractive index of water.

Solution



Since the total thickness is 18cm, then the thickness of water will be gotten from $n_{\text{water}} + n_{\text{flint glass}} + n_{\text{crown glass}} = 18 \text{ cm}$

$$x \text{ cm} + 8 \text{ cm} + 6 \text{ cm} = 18 \text{ cm}$$

$$x \text{ cm} = 18 \text{ cm} - 14 \text{ cm}$$

$$x = 4 \text{ cm} \quad (1)$$

Using the equation $D_{\text{water}} = x \left(1 - \frac{1}{n}\right)$

But total displacement $D = D_{\text{water}} + D_{\text{flint glass}} + D_{\text{crown glass}}$

$$D_{\text{water}} = x \left(1 - \frac{1}{n_w}\right)$$

$$D_{\text{flint glass}} = x_f \left(1 - \frac{1}{n_f}\right) = 8 \left(1 - \frac{1}{1.44}\right) = 2.44 \text{ cm} \quad (1)$$

$$D_{\text{crown glass}} = x_c \left(1 - \frac{1}{n_c}\right) = 6 \left(1 - \frac{1}{1.36}\right) = 1.5886 \text{ cm} \quad (2)$$

$$\text{Total displacement} = D_w + D_f + D_c$$

$$\Rightarrow D_w = 6 \text{ cm} - (2.44 + 1.5886) \text{ cm}$$

$$= 6 \text{ cm} - 4.0286$$

$$= 1.9714 \text{ cm} \quad (1)$$

$$D_w = x_w \left(1 - \frac{1}{n_w}\right)$$

$$n_w = \frac{x_w}{x_w - D_w} = \frac{4}{4 - 1.9714}$$

$$n_w = \frac{4}{2.0286} = 1.97$$

$$n_w = 1.97 \quad (1)$$

Ques 2.

A young lady looked at her face in front of a cosmetic 'concave' mirror, the erect image formed is 85 times the face of the lady. What will the focal length of the mirror be, if she held the mirror 45cm from her face.

$$u = 45, v = ? \quad m = +5.0 \quad \text{find } f.$$

$$= -5 + 45 = -225 \text{ cm}$$

$$\therefore \frac{1}{f} = \frac{1}{45} + \frac{1}{-225} = \frac{4}{225}$$

$$\therefore f = \frac{225}{4} = 56.25$$

③ find the angle of incident of a object placed at the bottom of a glass block 18 cm thick, displaced 4.4 cm upward. If the angle of refraction of an observer vertically above is 45° .

Soln
Take note that light rays travel from the object beneath the glass to the eye of the observer, hence light travels from glass to air.
from Snell's law $n_g \sin i_g = \frac{\sin i_a}{\sin r_a}$

But $\sin r_a = \frac{\text{real depth}}{\text{apparent depth}}$, $t = 18 \text{ cm}$, $d = 4.4 \text{ cm}$

$$\sin r_a = \frac{t}{t-d} = \frac{18}{18-4.4} = 1.324$$

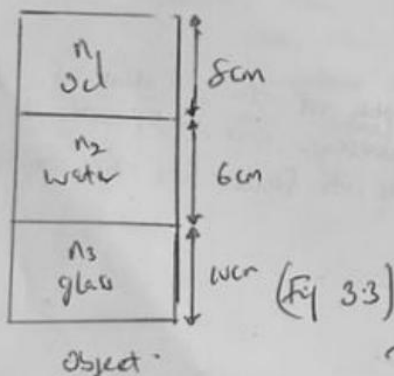
$$\sin r_a = 1.324$$

But $n_g \sin i_g = \frac{\sin r_a}{\sin r_a}$ but $r_a = 45^\circ$

$$\frac{1}{1.32} = \frac{\sin i_g}{\sin 45^\circ}$$

$$i_g = \sin^{-1} \left[\frac{\sin 45^\circ}{1.324} \right] = 32.28^\circ$$

④ Consider a film of oil 8 cm with refractive index $n=1.48$ floating on top of water 6 cm thick having refractive index $n=1.33$, on top of a glass block 10 cm with refractive index $n=1.62$, all contained in one medium. If an object is placed at the bottom of the glass block, calculate the apparent position of the object from above.



Soln

We will first obtain the displacement for the various medium series

$$D_{oil} = x_1 \left(1 - \frac{1}{n_1}\right) = 8 \left(1 - \frac{1}{1.48}\right) = 2.5946 \text{ cm}$$

$$D_{water} = x_2 \left(1 - \frac{1}{n_2}\right) = 6 \left(1 - \frac{1}{1.33}\right) = 1.4857 \text{ cm}$$

$$D_{glass} = x_3 \left(1 - \frac{1}{n_3}\right) = 10 \left(1 - \frac{1}{1.62}\right) = 3.827 \text{ cm}$$

$$\begin{aligned} \text{Total displacement} &= D_{oil} + D_{water} + D_{glass} \\ &= (2.5946 + 1.4857 + 3.827) \text{ cm} \\ &= 7.9073 \text{ cm} \end{aligned}$$

- ⑤ The image formed by an amusement park spherical mirror for anyone standing 2.5m in front of it is always upright and twice the person's height. What is the radius of curvature.

Soln

$$u = 2.5\text{m} \quad m = 2$$

$$\text{from } m = \frac{-v}{u}$$

$$2 = \frac{-v}{2.5}, \quad v = 2 \times 2.5 = -5\text{m}$$

$$\text{Using } \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{f} = \frac{2}{5} - \frac{1}{5} = \frac{1}{5}$$

$$\therefore f = 5\text{m}$$

$$\text{Recall } R = 2f = 2 \times 5\text{m} = 10\text{m}.$$

Assignment

- ① Describe the image of a candle flame located 40cm from a concave spherical mirror of radius 64cm. $\rightarrow$ Real, magnified and inverted
- ② What is the focal length of a convex spherical mirror which produces an image one-sixth the size of an object located 12cm from the mirror. 2.4cm
- ③ A mirror forms an erect image 30cm from the object and twice its height. (a) where must the mirror be situated $\rightarrow 10\text{cm}$ from object ✓
- ⑥ What is its radius of curvature $\rightarrow 40\text{cm}$ ✓
- ④ In a particular truck mirror, there is a warning 'object in the mirror appear to be more close than they appear' what kind of mirror must that be and why?
A mirror forms an erect image 40cm from the object and one-third height, where must the mirror be situated? What is its radius of curvature? If the object is real, is the mirror concave or convex?
 $f = -15\text{cm}, \quad r = -30\text{cm}$
- ⑤ An object 10.0cm tall is placed 40cm from a plane mirror. find ✓
 - ④ The distance from the object to the image $\rightarrow 80\text{cm}$ the mirror is plane, therefore the image distance must equal the distance of object
 - ⑤ The height of the image $\rightarrow 10.0\text{cm}$ $40 + 40 = 80\text{cm}$ ✓

REFRACTION THROUGH CURVED SURFACES (LENSES)

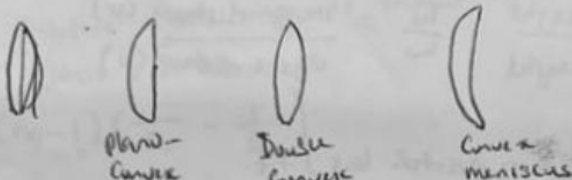
INTRODUCTION

A lens is an optical system with two refracting surfaces (most commonly made of glass but sometimes plastic or crystals). The refracting surfaces of a lens are close enough that we can neglect the distance between them (the thickness of the lens), we call this a thin lens.

~~Properties of lenses~~

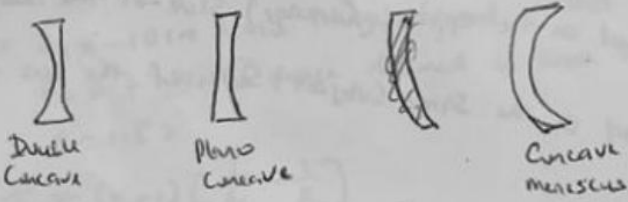
As we know in a spherical mirror, a lens with spherical geometry has for each of its surfaces a center of curvature, a radius of curvature, a focal point and a focal length. The focal point are at equal distances on either side of a thin lens. However for a spherical lens $f \neq R/2$ as it is for a spherical mirror. The curvature of the surface is measured for a within the solid so that:

- The convex lens surface has a real and therefore positive radius of curvature. The lens is thicker at the center than at its edges.
- A convex lens has a real focal point and therefore positive focal length.



Plano-convex Double convex Convex meniscus

- Concave lens surfaces have virtual and therefore negative radius of curvature, this lens is thin at the center and thicker at the edges.
- A concave lens has a virtual focal point and therefore negative focal length.



Double concave Plano concave Concave meniscus

N.B. All distances are measured from the pole of the lens and for a thin lens this is almost coincident with the center of the lens.

Geometric Properties of Lenses: $\rightarrow$ from Q.10

- Every lens has two curved surfaces (thus two centers of curvature (i.e. the center of a sphere of which the curve surface is a part)) C_1 and C_2 , hence two radii of curvature.
- Every lens has two focal points (i.e. points where parallel rays converge after passing through a converging lens or appears to diverge from after passing through a diverging lens) thus they also have two focal lengths.

the two principal axes of the lens are P_1 and P_2 .

Power of a Lens and Lens Equation
 The power of a lens is described in terms of either its focal length or the Optician unit of power of a lens is called diopter (D). The power of a lens is defined as:

$$\text{Power} = \frac{1}{\text{focal length (cm)}} \quad \text{or} \quad \frac{100}{\text{focal length (cm)}} = \frac{100 \text{ dioptre (D)}}{\text{focal length}}$$

The unit Dio is same as cm^{-1}

Convex lenses have positive power while Concave lenses have negative power. The power of a meniscus lens depends on which face of the lens it is more sharply curved.

For e.g., a ^{convex} lens with focal length of 10cm will have a power of +10D or while a concave lens with focal length of -5cm will have a power of -20D or the relationship between the power of lens object and image distance is called the lens equation.

If u = object distance, v = image distance, f = focal length then the power of a lens $P = \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ (lens equation).

$$\text{Magnification (M)} = \frac{\text{image height}}{\text{object height}} = \frac{h_i}{h_o} = \frac{\text{image distance (v)}}{\text{object distance (u)}}$$

NOTE

Converging lens (Sometimes called a positive lens) +f

Diverging lens (Sometimes called negative lens) -f

Object Distance (u)

+u when the object is in front of the lens (real object)

-u when the object is behind the lens (virtual object)

Image distance (v)

+v when the image is formed on the opposite (image) side of the lens from the object (real image)

-v when the image is formed on the same (object) side of the lens from the object (virtual image)

Image Orientation (i)

+m when the image is upright with respect to the object

-m when the image is inverted with respect to the object

In a combination of two or more lenses, the image formed by the first lens is taken as the object of the second lens (and so on).

If this image-object falls behind the second lens, it is referred to as a virtual object, and the object distance is taken to be

negative (-)

NOTE when we say the image is real it means the object and the image are on opposite sides of the lens $v > u$.

Concave lens always gives virtual, erect and diminished images.

Lens maker's equation

Any lens that is thicker at its center than at its edges is a converging lens with positive f , and any lens that is thicker at its ^{edges} center than at its center is a diverging lens with negative f . It can be proved by using the lens maker equation. This equation relates the shape of a lens (radius) R , its index of refraction n and the focal length f of the lens.

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

where $R_2 = \infty$ for a plane convex lens

For a concave lens, R_1 is negative since no image is formed on the side where R_2 is positive since image is formed on the side.

For air to glass, we have

$$\frac{1}{f} = (n_g - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Examples

(1) Suppose a concave lens is considered to have a single value of radius of curvature of 10cm and index of refraction of 1.52. What is the focal length of this lens?

Given $R_1 = -10\text{cm}$ (no image formed on that side)
 $R_2 = +10\text{cm}$ (image formed on same side of object)

$$n = 1.52$$

$$\text{Using } \frac{1}{f} = (n - 1) \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\frac{1}{f} = (1.52 - 1) \left[\frac{1}{-10} - \frac{1}{10} \right]$$

$$\frac{1}{f} = 0.102$$

$$\therefore f = -9.6\text{cm}$$

(2) A double convex lens made of glass ($n = 1.52$) has a radius of curvature of 50.0cm on the front side and 40cm on the back side. Find the power of the lens.

(4) A physics student used a diverging lens to examine an object. Determine the characteristics of the image of the object if it is placed 30cm in front of the lens of focal length 10cm

Soln Given $u = 30\text{cm}$
 $f = -10\text{cm}$ (Diverging lens)

Using $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$, $v = \frac{uf}{u-f}$

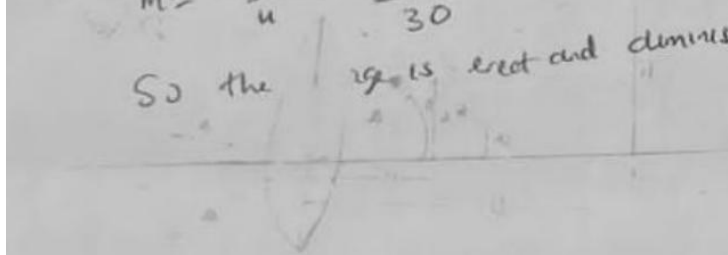
$$v = \left[\frac{30 \times -10}{30 - 10} \right] \text{cm} = -7.5\text{cm}$$

Since the image distance is negative it is virtual

The magnification

$$m = -\frac{v}{u} = -\frac{-7.5}{30} = 0.25$$

So the image is erect and diminished.



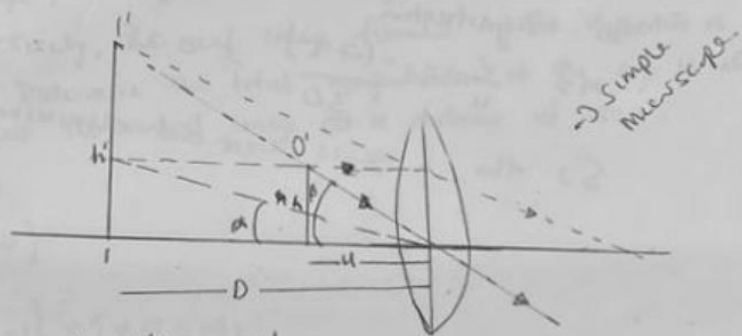
OPTICAL INSTRUMENTS

THE MICROSCOPE

A magnifying glass or microscope is an instrument used for looking at objects close to the eye and the image is usually formed at the least distance of distinct vision (D) from the eye.

THE SIMPLE MICROSCOPE

A convex lens of focal length f produces a magnified, erect and virtual image (I') of the object (O). The lens is adjusted so that the image is at the near point and so the image distance is the least distance of distinct vision D , about 25cm. A diagram of this is shown below.



from the diagram

$$\text{Angular Magnification, } M = \frac{\beta}{\alpha} = \frac{h'/D}{h/D} = \frac{h'}{h}$$

as long as the angles β and α are small

However, the linear magnification of the lens is

$$\frac{h'}{h} = \frac{v}{u} \text{ and since } \frac{1}{u} + \frac{1}{v} = \frac{1}{f} \text{ and } v = D$$

$$\frac{v}{u} = \frac{v}{f} - 1 \text{ therefore; } M = \frac{D}{f} - 1$$

when image is at infinity, and eye piece is close to the eye,

$$M = \frac{\beta}{\alpha} = \frac{h/f}{h/D} = \frac{D}{f}$$

In this case, M the object must be of principle $\frac{1}{f}$ This f

10) A physics student
Compound microscope

When we need greater magnification that we can get with the simple microscope, we usually use the compound microscope. The compound microscope uses several lenses to produce a highly magnified image of an object.

The magnifying power M of the instrument is given by the formula;
 $M = \frac{h_e}{h_o}$ and can be rewritten as $M = \frac{h_e}{h_o} = \frac{\beta}{\alpha}$ where $\frac{h_e}{h_o} = \frac{v}{f_e} - 1$
 $= \frac{D}{f_e} - 1$

Since $v = D$ and $\frac{h_e}{h} = \frac{v}{f_e} - 1$

$$M = \left(\frac{D}{f_e} - 1 \right) \left(\frac{v}{f_o} - 1 \right)$$

or

$$M = \frac{h_e}{h} = \frac{h}{h_o} \left(\frac{D}{f_e} - 1 \right) \left(\frac{v}{f_o} - 1 \right)$$

ex) ① An object 5cm high is placed 15cm in front of a convex lens of 10cm focal length. A second convex lens also of 10cm focal length is placed 48cm behind the first lens. find ① the image of the lens combination
 ② the magnification of the system and
 ③ the height of the final image

Solution

① the image of the first lens is found from

$$\frac{1}{v_1} = \frac{1}{f_1} - \frac{1}{u_1}$$

$$\frac{1}{v_1} = \frac{1}{10} - \frac{1}{15} = 0.0333 \text{ cm}^{-1}$$

$$v_1 = 30 \text{ cm}$$

The object distance for the second lens

$$u_2 = d - v_1 (v_1)$$

$$= 48 - 30$$

$$= 18 \text{ cm}$$

hence the final image is

$$\frac{1}{v_2} = \frac{1}{f_2} - \frac{1}{u_2} = \frac{1}{v_1} = \frac{1}{10} - \frac{1}{18} = 0.0444 \text{ cm}^{-1}$$

$$v_2 = 22.5 \text{ cm}$$

The magnification of each lens is

$$M_1 = \frac{v_1}{u_1} = -\frac{30}{15} = -2$$

$$M_2 = \frac{v_2}{u_2} = -\frac{22.5}{18} = -1.25$$

The final magnification of the system is

$$M = M_1 \cdot M_2 = (-2)(-1.25) = 2.5$$

⑤ The final image height of the system is:

$$h_2 = M \cdot h_1 = 2.5 \times 5 \\ = 12.5 \text{ cm}$$

②* In a compound microscope, the objective and eye-piece have focal lengths of 0.80 and 2.5 cm, respectively. The real image formed by the objective is 16 cm from the objective. Determine the total magnification if the eye is held close to eye piece and views the virtual image at a distance of 25 cm.

Soln - Using the formula

$$M = \left(\frac{v}{f_e} - 1 \right) \left(\frac{v}{f_o} - 1 \right) \\ = \left(\frac{25}{2.5} - 1 \right) \left(\frac{16}{0.8} - 1 \right) = 9 \times 19 = 171 \\ = 1.71 \times 10^2$$

Telescopes

telescope is used to view objects of large distances whereas the microscope is used to view small objects at close distance. Another difference is that, many telescopes use a curved mirror, not a lens, as an objective.

Astronomical telescope

The astronomical telescope is ^{an} example of combination of two converging lenses, the objective lens and the eye lens. However, the objective is a large converging lens of long focal length because the object is very far away. The astronomical telescope is not very good for viewing distant objects on the surface of the earth because they are all inverted. The astronomical telescope can be converted to a

Terrestrial telescope by placing a third converging or erecting lens of focal length f , between the objective and eye piece. The length or distance between objective lens and eye piece of a terrestrial telescope is $L = f_o + 4f$

$$\text{Angular magnification} = \frac{\text{Angle subtended by the eye (B)}}{\text{Angle subtended by the object at unaided eye (A)}}$$

$$= \frac{\text{focal length of objective (f}_o\text{)}}{\text{focal length of eye piece (f}_e\text{)}} = \frac{\text{or objective diameter}}{\text{eye piece diameter}}$$

GALILEAN TELESCOPE

It consists of a converging lens of large focal length for the objective and diverging lens of short focal length for the eye piece. The advantage of Galilean telescope is that it gives erect image.

Reflecting Telescopes
Terrestrial Telescopes

} Write short notes on this.

LASER
 Laser is a device that emits light through a process of optical amplification based on the stimulated emission of electromagnetic radiation.

A laser differs from other sources of light because it emits light coherently, and this coherent light produced by a laser differs from ordinary light in that it is made up of waves of the same wavelength and all in phase (i.e. step with each other). Ordinary light contains many different wavelengths and phase relations.

Characteristics of Lasers

The physical size of a laser depends on the materials used for light emission, on its power output, and on whether the light is ~~emitted~~ emitted in pulses or as a steady beam. In some lasers, a gas or liquid is used as the emitting medium.

Lasers focus between infrared and ultraviolet in the spectrum mainly in the visible light spectrum.

Properties of laser light are:

Monochromaticity (The same color)

Coherence (all of the light waves are in phase both spatially and temporally)

Collimation (all rays are parallel to each other and do not diverge significantly even over long distances.)

COMPARISON OF LASERS

Application of Lasers

Lasers have been used in industry for cutting and welding metals and other materials as well as welding and soldering and for inspecting optical equipment.

In medicine, they have been used in Surgical Operations or Laser Surgery and various skin treatments.

They are used in CDs and DVDs read and written and are employed in laser printers and bar-code scanners.

They are used in communications, both in fiber Optics and in some Space and open-air communications.

Lasers have been used in a number of areas of Scientific research, and have opened a new field of Scientific research, nonlinear Optics, which is concerned with the study of such phenomena as the frequency doubling of coherent light by certain crystals.

One important result of laser research is the development of lasers that can be tuned to emit light over a range of frequencies, instead of producing light only a single frequency.

only a single frequency.